

37.3 Poisson and multinomial

A contingency table is a table of counts, so a Poisson log-linear model can be fitted to it. But the counts usually come from a survey with a fixed sample size, or from strata with fixed totals, and then they are not independent Poisson variables at all: they are multinomial. The two descriptions lead to the same fit. That is why software for Poisson regression is also software for contingency tables, and it is the bridge to the baseline-category logit models of [Chapter 36](#).

37.3.1 Conditioning on the total

Theorem 37.3.1 — Poisson counts conditioned on their total

- (a) Let $Y_1, \dots, Y_c$ be independent, Y_j Poisson with mean $\mu_j > 0$, and $N = \sum_j Y_j$. Then N is Poisson with mean $\mu_+ = \sum_j \mu_j$, and conditionally on $N = n$ the vector $(Y_1, \dots, Y_c)$ is multinomial with n trials and probabilities $\pi_j = \mu_j / \mu_+$.
- (b) Suppose the cells are indexed by a stratum $s = 1, \dots, S$ and a category $j = 1, \dots, c$, the counts Y_{sj} are independent Poisson with means μ_{sj} , and the log-linear model is

$$\log \mu_{sj} = \alpha_s + \mathbf{z}_{sj}^\top \boldsymbol{\gamma}, \quad (37.3.1)$$

with $\boldsymbol{\alpha} \in \mathbb{R}^S$ unrestricted: that is, the model contains a free parameter for every stratum. Then the Poisson log-likelihood splits as

$$\ell(\boldsymbol{\alpha}, \boldsymbol{\gamma}) = \ell_1(\mu_{1+}, \dots, \mu_{S+}) + \ell_2(\boldsymbol{\gamma}) + \text{constant},$$

where ℓ_1 is the log-likelihood of S independent Poisson totals and ℓ_2 is the log-likelihood of S independent multinomial samples with category probabilities

$$\pi_{sj}(\boldsymbol{\gamma}) = \frac{\exp(\mathbf{z}_{sj}^\top \boldsymbol{\gamma})}{\sum_{l=1}^c \exp(\mathbf{z}_{sl}^\top \boldsymbol{\gamma})}, \quad (37.3.2)$$

and the two pieces have no parameters in common.

- (c) Consequently the maximum likelihood estimate $\hat{\boldsymbol{\gamma}}$ from the Poisson fit is the maximum likelihood estimate from the multinomial fit; the fitted cell counts are $\hat{\mu}_{sj} = n_s \hat{\pi}_{sj}$ with $n_s = \sum_j y_{sj}$; the observed information for $\boldsymbol{\gamma}$ is the same in the two models; and deviance differences between models that all contain the stratum parameters agree. The stratum parameters $\hat{\alpha}_s$ themselves have no multinomial counterpart: they are determined by $\hat{\mu}_{s+} = n_s$.

Proof.

- (a) Sums of independent Poisson variables are Poisson with the sum of the means (convolve the mass functions, or multiply moment generating functions), so $\Pr(N =$

$n) = e^{-\mu_+} \mu_+^n / n!$. For nonnegative integers y_j with $\sum_j y_j = n$,

$$\begin{aligned} \Pr(Y_1 = y_1, \dots, Y_c = y_c \mid N = n) &= \frac{\prod_j e^{-\mu_j} \mu_j^{y_j} / y_j!}{e^{-\mu_+} \mu_+^n / n!} \\ &= \frac{n!}{\prod_j y_j!} \prod_j \pi_j^{y_j}, \end{aligned}$$

because $\sum_j \mu_j = \mu_+$ makes the exponential factors cancel and $\prod_j \mu_j^{y_j} = \mu_+^n \prod_j \pi_j^{y_j}$. That is the multinomial mass function.

- (b) Write $\mu_{s+} = \sum_j \mu_{sj}$ and $\pi_{sj} = \mu_{sj} / \mu_{s+}$. Under [Eq. 37.3.1](#), $\mu_{s+} = e^{\alpha_s} \sum_l \exp(\mathbf{z}_{sl}^\top \boldsymbol{\gamma})$, and the ratio π_{sj} is [Eq. 37.3.2](#), free of $\boldsymbol{\alpha}$. Substituting $\mu_{sj} = \mu_{s+} \pi_{sj}$ in the Poisson log-likelihood,

$$\begin{aligned} \ell &= \sum_{s,j} \{y_{sj} \log \mu_{sj} - \mu_{sj}\} - \sum_{s,j} \log y_{sj}! \\ &= \sum_s \{n_s \log \mu_{s+} - \mu_{s+} - \log n_s!\} \\ &\quad + \sum_s \left\{ \log n_s! - \sum_j \log y_{sj}! + \sum_j y_{sj} \log \pi_{sj} \right\}, \end{aligned}$$

which is the stated split, the first sum being ℓ_1 and the second ℓ_2 plus the multinomial coefficients. For fixed $\boldsymbol{\gamma}$, the map $\boldsymbol{\alpha} \mapsto (\mu_{1+}, \dots, \mu_{S+})$ is a bijection from $\mathbb{R}^S$ onto $(0, \infty)^S$, so the parameters $(\mu_{1+}, \dots, \mu_{S+})$ and $\boldsymbol{\gamma}$ vary independently.

- (c) The two pieces share no parameters and their ranges are unconstrained, so the joint maximum is attained by maximizing each separately: ℓ_1 gives $\hat{\mu}_{s+} = n_s$ and ℓ_2 is the multinomial problem, giving $\hat{\boldsymbol{\gamma}}$ and $\hat{\mu}_{sj} = n_s \hat{\pi}_{sj}$. The information for $\boldsymbol{\gamma}$ is minus the second derivative of ℓ_2 , and differences of maximized log-likelihoods between models sharing the stratum parameters lose the common ℓ_1 .

■

Part (c) is the practical statement: *fit the table as independent Poisson counts, and provided the model contains a free parameter for every total the design fixed, you get the multinomial answer.* The converse matters as much: if the design fixed the row totals and the model omits the row parameters, the two fits differ and the Poisson one answers a question the design cannot. The proof used s only to group cells, so any partition into blocks with fixed totals works the same way.

37.3.2 Independence in a two-way table

Take $S = r$ rows and c columns, and let $\mathbf{z}_{ij}$ consist of the column indicators alone, so that [Eq. 37.3.1](#) reads

$$\log \mu_{ij} = \lambda + \lambda_i^R + \lambda_j^C, \quad (37.3.3)$$

the log-linear model with row and column main effects and no interaction. The parameters are redundant by two, and any of the usual side conditions removes the redundancy without changing the fit (Theorem 6.7.1).

Corollary 37.3.2 — The independence model

Under Eq. 37.3.3 the conditional distribution of the column given the row does not depend on the row, so the two classifications are independent. The maximum likelihood fitted values are

$$\hat{\mu}_{ij} = \frac{y_{i+} y_{+j}}{n}, \quad y_{i+} = \sum_j y_{ij}, \quad y_{+j} = \sum_i y_{ij}, \quad n = y_{++}, \quad (37.3.4)$$

the deviance is $G^2 = 2 \sum_{i,j} y_{ij} \log(y_{ij}/\hat{\mu}_{ij})$, the Pearson statistic is $X^2 = \sum_{i,j} (y_{ij} - \hat{\mu}_{ij})^2 / \hat{\mu}_{ij}$, and both have $(r-1)(c-1)$ residual degrees of freedom — and, when the expected counts are large, both are asymptotically χ^2 on that many degrees of freedom (Theorem 34.5.5).

Proof. With $\mathbf{z}_{ij}$ free of i , Eq. 37.3.2 gives $\pi_{ij} = \pi_j$, which is independence. By Theorem 37.1.2(b) applied to the row and to the column indicators, $\hat{\mu}_{i+} = y_{i+}$ and $\hat{\mu}_{+j} = y_{+j}$. Under Eq. 37.3.3, $\hat{\mu}_{ij} = \hat{\mu}_{i+} \hat{\mu}_{+j} / \hat{\mu}_{++}$, because $\hat{\mu}_{ij} = e^{\lambda a_i b_j}$ factorizes and summing gives $\hat{\mu}_{i+} \hat{\mu}_{+j} = e^{2\lambda a_i b_j} \sum_l b_l \sum_k a_k = \hat{\mu}_{ij} \hat{\mu}_{++}$. Substituting the matched margins gives the fit Eq. 37.3.4. The model has $1 + (r-1) + (c-1)$ free parameters against rc cells, leaving $rc - r - c + 1 = (r-1)(c-1)$. The deviance formula is Eq. 37.1.5 with the second term vanishing by Theorem 37.1.2(a). ■

The fitted values Eq. 37.3.4 are the expected counts of the elementary chi-squared test for independence, available since Pearson (1900); what the log-linear framework adds is everything around it.

37.3.3 Association is interaction on the log scale

Adding an interaction term to Eq. 37.3.3 gives the saturated model

$$\log \mu_{ij} = \lambda + \lambda_i^R + \lambda_j^C + \lambda_{ij}^{RC}, \quad (37.3.5)$$

with rc free parameters and $\hat{\mu}_{ij} = y_{ij}$.

Proposition 37.3.3 — Interaction parameters are log odds ratios

In Eq. 37.3.5, for any rows $i \neq k$ and columns $j \neq l$,

$$\lambda_{ij}^{RC} + \lambda_{kl}^{RC} - \lambda_{il}^{RC} - \lambda_{kj}^{RC} = \log \frac{\mu_{ij} \mu_{kl}}{\mu_{il} \mu_{kj}},$$

which does not depend on the side conditions (Theorem 8.4.2) used to identify the parameters. Under the multinomial reading of Theorem 37.3.1 the right-hand side is the log odds ratio $\log\{\pi_{ij}\pi_{kl}/(\pi_{il}\pi_{kj})\}$. The model Eq. 37.3.3 holds if and only if every such quantity is zero.

Proof. In the four terms of Eq. 37.3.5 the constant, the row effects and the column effects each appear twice with a plus and twice with a minus sign, so they cancel, leaving the stated combination equal to $\log \mu_{ij} + \log \mu_{kl} - \log \mu_{il} - \log \mu_{kj}$. The cancellation uses nothing about the parameterization, so any two parameterizations of the same fitted means agree. Dividing every μ by μ_{++} leaves the ratio unchanged, giving the multinomial form; and all such quantities vanish exactly when $\log \mu_{ij}$ is additive in i and j , which is Eq. 37.3.3. ■

The combination in Proposition 37.3.3 is the tetrad difference of Definition 16.2.1 applied to $\log \mu_{ij}$ instead of to a cell mean, so the apparatus of Chapter 16 transfers: interaction contrasts survive reparameterization (Proposition 16.2.1) and main effects do not. The difference is that there the scale was chosen for convenience and could be changed by a transformation (Proposition 16.2.2), whereas here the log scale is forced by the link and “no interaction” means “independence”.

Example. (Education and party identification)

The 1996 American National Election Studies sample distributed with `statsmodels` records 944 respondents. Cross-classify them by education in four levels (no diploma, high school graduate, some college, degree) and by party identification collapsed to three, with column totals 380, 239 and 325.

Fitting the independence model Corollary 37.3.2 gives $G^2 = 20.541$ and $X^2 = 19.338$ on six degrees of freedom. The smallest expected count is 16.46, so the χ^2 calibration is safe, and the p -value is 0.0022: independence is rejected. The largest Pearson residual $(y_{ij} - \hat{\mu}_{ij})/\sqrt{\hat{\mu}_{ij}}$ is -2.62 , at Republicans without a diploma, where 10 were observed against 22.38 expected; the largest positive one, 2.31 , is at Democrats in the same row.

The alternative need not be saturated. Education is ordered, so give it scores $u_i = 1, 2, 3, 4$ and fit $\log \mu_{ij} = \lambda + \lambda_i^R + \lambda_j^C + u_i \delta_j$, spending two parameters instead of six on the association. Its deviance is 7.367 on four degrees of freedom, p -value 0.1177: this model fits, and the drop from independence is 13.174 on two degrees of freedom, $p = 0.00138$.

Now read the same three fits as multinomial logit models for party given education (Definition 36.1.1), with Democrat as the baseline: independence becomes the intercept-only logit model, the score model a logit model linear in u_i , the saturated model a logit model with a free effect per education level. Each pair gives identical fitted counts, and the association parameters δ_j are exactly the education slopes of the logit model, 0.1513 for Independent against Democrat and 0.2748 for Republican against Democrat: each step up the education scale multiplies the odds of Republican rather than Democrat by 1.3163. In the saturated model the interaction contrast comparing the degree row with the no-diploma row and the Republican column with the Democrat column is 1.3658, so those odds are 3.919 times as large among respondents with a degree as among those without a diploma.

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
```

```

anes = sm.datasets.anes96.load_pandas().data
party = pd.cut(anes["PID"], [-0.5, 1.5, 4.5, 6.5],
               labels=["Democrat", "Independent", "Republican"])
education = pd.cut(anes["educ"], [0.5, 2.5, 3.5, 4.5, 7.5],
                  labels=["no diploma", "high school", "some college", "degree"])
table = pd.crosstab(education, party)
print(table)

```

[Run this code online](#)

```

def poisson_mle(X, y, tol=1e-12, maxit=80):
    """Fisher scoring for log mu = X beta."""
    beta = np.zeros(X.shape[1])
    beta[0] = np.log(y.mean())
    for _ in range(maxit):
        mu = np.exp(X @ beta)
        XW = X * mu[:, None]
        step = np.linalg.solve(X.T @ XW, XW.T @ (X @ beta + (y - mu) / mu))
        if np.max(np.abs(step - beta)) < tol:
            return step
        beta = step
    return beta

def dummies(levels, k):
    """Indicator columns for levels 1, ..., k-1 of a factor taking the values 0..k-1."""
    return np.column_stack([(levels == j).astype(float) for j in range(1, k)])

counts = table.to_numpy(float)
r, c = counts.shape
score = np.arange(1.0, r + 1) # education score u_i = 1, 2, 3, 4
rows, cols = np.divmod(np.arange(counts.size), c)
y = counts.ravel()
main = np.column_stack([np.ones(counts.size), dummies(rows, r), dummies(cols, c)])
linear = np.column_stack([main, score[rows][:, None] * dummies(cols, c)])
saturated = np.column_stack([main] + [dummies(rows, r)[:i, [i]] * dummies(cols, c)[:i, [j]]
                                     for i in range(r - 1) for j in range(c - 1)])

fits = {}
for name, X in [("independence", main), ("linear", linear), ("saturated", saturated)]:
    beta = poisson_mle(X, y)
    fits[name] = (beta, np.exp(X @ beta).reshape(counts.shape))

print("independence fit:\n", np.round(fits["independence"][1], 2))
print("row totals match:", np.allclose(fits["independence"][1].sum(1), counts.sum(1)))
print("column totals match:", np.allclose(fits["independence"][1].sum(0), counts.sum(0)))

```

[Run this code online](#)

```

from scipy import stats

```

```
def deviance(observed, fitted):
    return 2 * np.sum(observed * np.log(observed / fitted))

for name, dfree in [("independence", 6), ("linear", 4)]:
    dev = deviance(counts, fits[name][1])
    pear = np.sum((counts - fits[name][1]) ** 2 / fits[name][1])
    print(f"{name:13s} G2 = {dev:6.3f} X2 = {pear:6.3f} df = {dfree} "
          f"p = {stats.chi2.sf(dev, dfree):.4f}")
```

[Run this code online](#)

The next listing fits the three multinomial logit models directly, by Fisher scoring on the multinomial likelihood, and confirms that each reproduces the fitted counts of its log-linear partner.

```
def multinomial_logit(counts, design, tol=1e-12, maxit=100):
    """Baseline-category logits:  $\log(\pi_{i-k} / \pi_{i0}) = \text{design}_i \cdot \gamma_k$ ,  $k = 1..c-1$ ."""
    r, c = counts.shape
    q = design.shape[1]
    gamma = np.zeros((c - 1) * q)
    total = counts.sum(1)
    for _ in range(maxit):
        eta = np.column_stack([np.zeros(r)] +
                               [design @ gamma[k * q:(k + 1) * q] for k in range(c - 1)])
        pi = np.exp(eta) / np.exp(eta).sum(1, keepdims=True)
        mu = total[:, None] * pi
        u = np.concatenate([design.T @ (counts[:, k + 1] - mu[:, k + 1])
                             for k in range(c - 1)])
        info = np.zeros((len(gamma), len(gamma)))
        for k in range(c - 1):
            for l in range(c - 1):
                w = total * pi[:, k + 1] * ((k == l) - pi[:, l + 1])
                info[k * q:(k + 1) * q, l * q:(l + 1) * q] = design.T @ (design * w
                                                                          [:, None])
        step = np.linalg.solve(info, u)
        gamma = gamma + step
        if np.max(np.abs(step)) < tol:
            break
    return gamma, mu

designs = {"independence": np.ones((r, 1)),
          "linear": np.column_stack([np.ones(r), score]),
          "saturated": np.column_stack([np.ones(r), dummies(np.arange(r), r)])}
for name, design in designs.items():
    gamma, mu = multinomial_logit(counts, design)
    print(f"{name:13s} largest difference in fitted counts: "
          f"{np.abs(mu - fits[name][1]).max():.2e}")
```

[Run this code online](#)

37.3.4 More than two classifications

With three classifications the same ideas give a family of models, one for each independence structure. Writing the saturated model as

$$\begin{aligned} \log \mu_{ijk} = & \lambda + \lambda_i^A + \lambda_j^B + \lambda_k^C \\ & + \lambda_{ij}^{AB} + \lambda_{ik}^{AC} + \lambda_{jk}^{BC} + \lambda_{ijk}^{ABC}, \end{aligned}$$

the standard submodels delete terms, always a higher-order term before any lower-order term it contains (a **hierarchical** model). Dropping all two-factor terms gives mutual independence; keeping λ^{AC} and λ^{BC} makes A and B conditionally independent given C ; dropping only λ^{ABC} gives **homogeneous association**, in which the AB odds ratios are the same at every level of C ([Exercise \(C2\)](#), [Exercise \(B2\)](#)).

Theorem 37.1.2 makes the fitting transparent: the likelihood equations equate observed and fitted counts in exactly the margins spanned by the model's terms. For mutual independence those are the one-way margins and the fit is a product of proportions; for homogeneous association they are all three two-way margins, which has no closed form and is what iterative fitting is for.

Remark (Graphical models). Draw a vertex for each classification and an edge for each two-factor term the model retains. For the hierarchical models whose structure that graph captures, conditional independence given a separating set can be read off the picture: these are the **graphical log-linear models** of Darroch, Lauritzen and Speed (1980), the discrete ancestors of graphical models generally. They also say when collapsing over a variable leaves an association unchanged, which is never safe to assume: the AB association survives collapsing over C only when C is conditionally independent of A , or of B , given the other (Bishop, Fienberg and Holland, 1975). Otherwise marginal and conditional associations may differ in size or in sign, as [Section 14.5](#) found for partial correlations.

37.3.5 Exercises

37.3.5.1 A. Check your understanding

Exercise A1.

A three-way table is $3 \times 4 \times 2$. How many cells does it have, how many free parameters does the homogeneous association model have, and what are the residual degrees of freedom?

Exercise A2.

Verify from [Theorem 37.3.1\(a\)](#) that the conditional distribution of Y_1 given $N = n$ is binomial with n trials and probability μ_1/μ_+ , and use it to explain why a Poisson model for two counts is the same as a binomial model for one of them given the total.

37.3.5.2 B. Practice

Exercise B1.

For the 2×2 table with entries $y_{11}, y_{12}, y_{21}, y_{22}$, compute the independence fit [Eq. 37.3.4](#) by hand and show that $y_{ij} - \hat{\mu}_{ij} = \pm(y_{11}y_{22} - y_{12}y_{21})/n$ with alternating signs. Deduce that $X^2 = n(y_{11}y_{22} - y_{12}y_{21})^2 / (y_{1+}y_{2+}y_{+1}y_{+2})$.

Exercise B2.

For the conditional independence model $\log \mu_{ijk} = \lambda + \lambda_i^A + \lambda_j^B + \lambda_k^C + \lambda_{ik}^{AC} + \lambda_{jk}^{BC}$, show that the likelihood equations equate the AC and BC margins, that $\hat{\mu}_{ijk} = y_{i+k}y_{+jk}/y_{++k}$, and that the residual degrees of freedom are $(r-1)(c-1)\ell$ for an $r \times c \times \ell$ table.

Solution. The terms in the model span the indicators of the AC and BC cells, so Theorem 37.1.2 equates those margins. The model says $\hat{\mu}_{ijk} = f_{ik}g_{jk}$, so within each level k it is the independence model of [Corollary 37.3.2](#) applied to the k th layer, whose fit is $y_{i+k}y_{+jk}/y_{++k}$. Each layer contributes $(r-1)(c-1)$ residual degrees of freedom, and there are ℓ layers.

Exercise B3.

In a $2 \times c \times \ell$ table with A binary, show that the homogeneous association model implies that the logit of $\Pr(A = 1 \mid B = j, C = k)$ is additive in j and k , and identify the logistic regression that corresponds to it. Which log-linear terms determine the logistic intercept, which the slopes, and which do not appear at all?

Solution. The logit is $\log(\mu_{1jk}/\mu_{2jk})$. Substituting the model, the terms $\lambda, \lambda_j^B, \lambda_k^C$ and λ_{jk}^{BC} cancel, leaving $(\lambda_1^A - \lambda_2^A) + (\lambda_{1j}^{AB} - \lambda_{2j}^{AB}) + (\lambda_{1k}^{AC} - \lambda_{2k}^{AC})$, which is additive: it is the main-effects logistic regression of A on B and C . The A main effect gives the intercept and the AB, AC terms the slopes; λ^{BC} , the association between the explanatory factors, does not appear.

37.3.5.3 C. Going deeper**Exercise C1.**

The model of [Example \(Education and party identification\)](#) with row scores, $\log \mu_{ij} = \lambda + \lambda_i^R + \lambda_j^C + u_i\delta_j$, is called a row-effects model. Show that its local log odds ratios $\log\{\mu_{ij}\mu_{i+1,j+1}/(\mu_{i,j+1}\mu_{i+1,j})\}$ equal $(u_{i+1} - u_i)(\delta_{j+1} - \delta_j)$. Then show that giving *both* classifications scores, with association term $\theta u_i v_j$, makes every local log odds ratio $\theta(u_{i+1} - u_i)(v_{j+1} - v_j)$ — the **uniform association** model — and count its parameters.

Exercise C2.

For a three-way table, show that the likelihood equations of the mutual independence model are $\hat{\mu}_{i++} = y_{i++}$, $\hat{\mu}_{+j+} = y_{+j+}$ and $\hat{\mu}_{++k} = y_{++k}$, and solve them to get $\hat{\mu}_{ijk} = y_{i++}y_{+j+}y_{++k}/n^2$. Explain why the homogeneous association model has no such closed form.